Skill Pilot AI

LLM Powered Career Coach

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Abstract

Overview

SkillPilotAI combines resume parsing, an LLM-based career coach, and personalized job/course recommendations into one platform—helping users turn resume data into actionable career guidance.

Methodology

1. Parse resume → build user profile (skills, education, experience)
2. Use an LLM to generate guidance (roles, learning path, roadmap) from the profile
3. Match jobs and recommend courses by comparing requirements vs. user skills

Applications

Career guidance chatbot, skill-gap analysis, job matching with missing-skill insights, and personalized course recommendations for upskilling.

Problem Statement & Analytical Study

Identified Gaps in Career Guidance:

Most students and early professionals lack access to consistent, personalized career mentorship. Existing solutions are fragmented—resume review tools, job boards, and course platforms operate separately—making it hard to identify skill gaps, choose the right career path, and take focused action.

Proposed Solution:

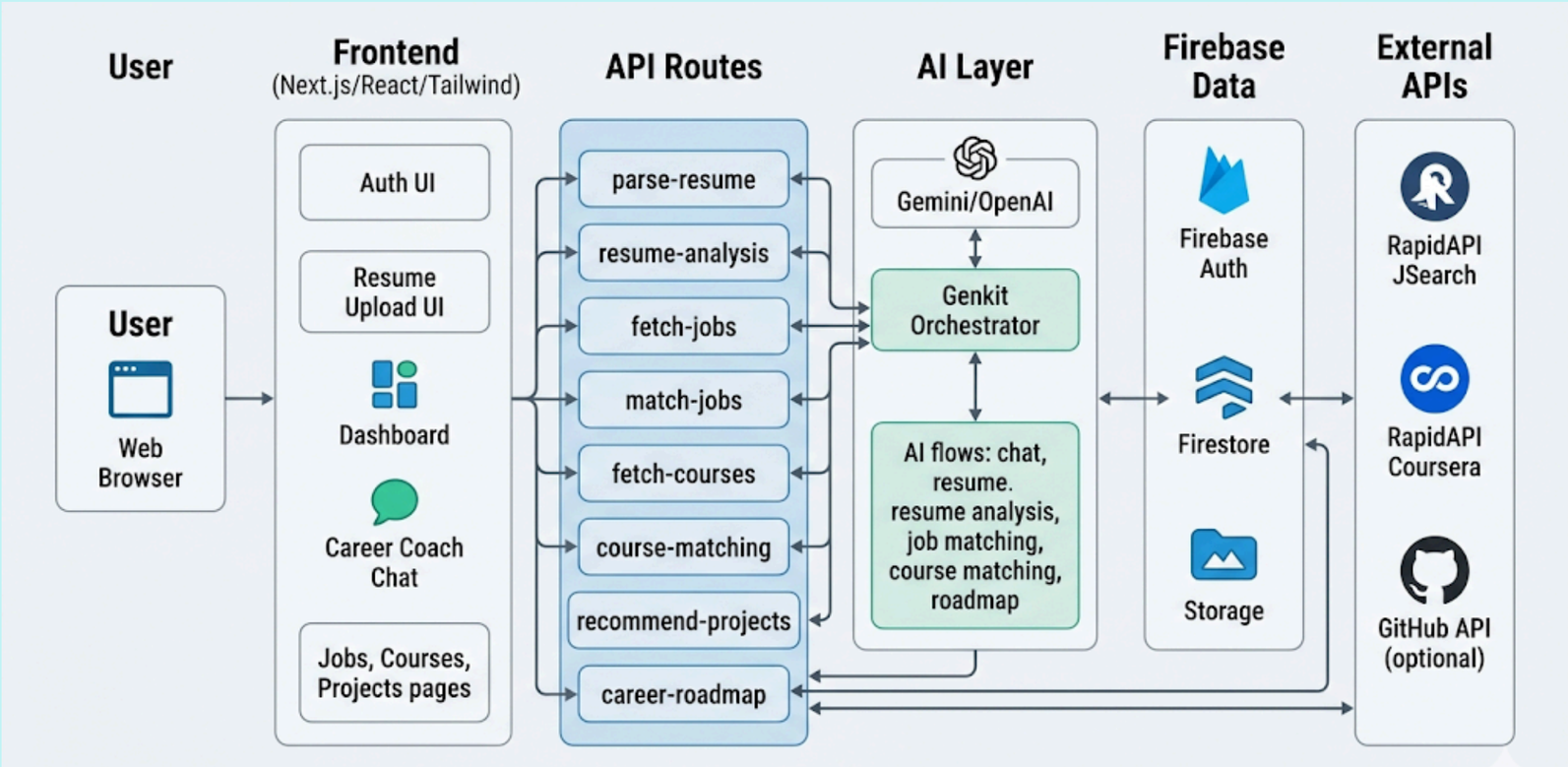
SkillPilotAI consolidates the full career-growth pipeline:

- Resume upload and parsing to extract skills, education, and experience
- Automatic profile creation and storage for reusable context
- LLM-based career coach for role guidance, Q&A, and roadmap generation
- Job discovery via external job APIs and AI-based job–skill matching
- Skill-gap detection with targeted course recommendations for upskilling

Key Differentiator:

A single integrated platform where resume-derived context powers all modules (chat, matching, gap analysis, and learning recommendations), enabling more personalized and actionable guidance than isolated tools.

System Architecture & Project Scope



ML Model Results & Predictions

Sample Output Metrics (Resume → Career guidance Pipeline)

Baseline: Metrics summarize end-to-end performance on sample resumes and generated recommendations. Higher parsing and skill extraction improves downstream matching, course ranking, and roadmap personalization.

ML Model Results & Predictions

Evaluation Engine

Evaluates the end-to-end career guidance pipeline (resume → profile → recommendations) using automated checks and user-feedback signals.
Key metrics compound:

Personalization Optimization

Uses resume-derived skills + goals + interactions to refine results across chat, job matching, and course ranking.
Constraints include:

- Minimum relevance threshold for recommendations (avoid generic outputs).
- Skill-gap prioritization (high-impact missing skills first)
- Diversity across roles/resources (avoid repetitive suggestions)

Conclusion

SkillPilotAI shows how an LLM-based system can turn raw resume information into meaningful, personalized career guidance. By combining resume parsing, automated profile creation, and an AI career coach chatbot, the platform helps users understand their current strengths, identify skill gaps, and receive clear recommendations for suitable roles and learning paths.

By integrating job matching, course recommendations, and roadmap generation within one application, SkillPilotAI reduces the need to rely on multiple disconnected tools and makes career planning more actionable. With future enhancements such as improved evaluation tracking, broader data sources, and stronger document support (e.g., PDF parsing), the platform can become a scalable and reliable career companion for students, job seekers, and training programs.

ML Model Results & Predictions

Future Enhancements:

- Add PDF resume parsing (OCR + layout-aware extraction) and better multi-column resume support
- Improve personalization with long-term memory (career goals, preferences) and feedback-driven refinement
- Real-time recommendation updates using user interactions (clicks, saves, completions)
- Expand job and course sources (multi-provider aggregation, deduplication, ranking improvements)

References:

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